

Game Theory Does Not Always Help: The Case of Statistical Multi-Party Coin Tossing

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Abstract. The study of coin-tossing protocols lies at the intersection of cryptography and game theory, where parties with potentially conflicting interests aim to jointly generate an unbiased random bit. Classical cryptographic results establish that strong fairness is achievable with an honest majority in the statistical setting, but impossible with a dishonest majority. In parallel, game-theoretic approaches [TCC 2018, Eurocrypt 2022 & CRYPTO 2024] have demonstrated that weaker equilibrium-based fairness guarantees can sometimes circumvent cryptographic lower bounds, raising the question of whether such techniques can overcome impossibility in the statistical regime.

In this work, we answer this question negatively.

- We show that for n parties and $t \geq n/2$ corruptions, there exists no statistically secure, game-theoretic coin-tossing protocol, even assuming broadcast and bounded round complexity—with the sole exception of the case $n = 4$ where all but one party share the same preference. This complements known feasibility results for $t < n/2$, thereby completing the statistical feasibility landscape.
- We further consider the setting without broadcast. Here we prove that no computationally secure game-theoretic coin-tossing protocol exists for $t \geq n/3$ and polynomial-round complexity, except for the corner case $n = 6$ where all but one party share the same preference.

To establish these results, we refine existing frameworks for game-theoretic fairness to capture both broadcast and point-to-point communication models.

Together, our results establish the boundaries of game-theoretic fairness in multi-party coin tossing: while it extends feasibility in the computational setting, it offers no advantage in the statistical setting once an honest majority is lost.

1 Introduction

Both game theory and cryptography examine the interaction of parties with potentially conflicting interests. Cryptography defines worst-case guarantees, such as privacy, correctness and fairness, that must hold even against fully malicious coalitions. Game theory instead studies rational adversaries, asking whether protocols with incentive structures form equilibria (e.g., Nash or sequential equilibrium) in which deviations are *unprofitable* rather than *impossible*. With the advent of decentralized technologies, an emerging line of work has highlighted how the two fields can mutually inform and strengthen each other.

On the game-theoretic side, cryptographic techniques such as secure multi-party computation (MPC) have been used to replace trusted parties. For example, Dodis and Rabin [22] demonstrated how MPC can help to eliminate the need for a trusted mediator in achieving certain game-theoretic concepts, such as correlated equilibria. Ferreira and Weinberg [24] showed how to use cryptographic commitments to circumvent known impossibility results in auctions by relaxing some of the requirements (instead of ensuring that there exists no profitable deviation, the work ensures that no auctioneer *can find* one without breaking cryptographic assumptions). Timed cryptographic primitives such as time-lock puzzles [38,34] are known to facilitate dishonest majority sealed-bid auctions without a trusted auctioneer.

On the cryptographic side, game-theoretic techniques and notions of security—especially fairness—have been used to circumvent classical impossibility results [16,43,41]. An exciting line of work focuses specifically on the *coin-tossing* functionality. Informally, n parties, each with its own source of randomness, wish to jointly toss a coin in a way that the output is unbiased and unpredictable. Coin-tossing has a long history in cryptography and distributed computing, with applications ranging from leader election to consensus protocols [2,7,6,10,30,29,36,12,19,39,23,21,3,9,1,40,33,35,32,20,26,27,28].

At the heart of this literature are two classical results. Cleve’s lower bound [17] showed that a two-party coin toss with one corruption is impossible to achieve—a corrupt party can bias the honest party’s outcome by a non-negligible amount. In contrast, Blum’s celebrated coin-tossing protocol [11] achieves a weaker but still meaningful guarantee: neither party can bias the outcome towards its *preferred* bit. This distinction is particularly interesting in the game-theoretic setting, where protecting against rational parties seeking to improve utility (rather than arbitrary malicious deviation) may be sufficient [42,15,14,13,4,5].

More recently, Chung, Guo, Lin, Pass, and Shi [16] explored the feasibility of n -party coin toss under such game-theoretic notions in the presence of a dishonest majority. In more detail, assuming a public preference profile of parties, Chung et al. [16] constructed protocols which satisfy the game-theoretic notion of fairness dubbed *Cooperative-Strategy-Proofness* (*CSP-fairness*) for an n -party setting with dishonest majority where all parties, except one, prefer the same bit b . On

⁶ The parameter regime has a more fine-grained dependence on the number of parties and their preference profiles. We refer the reader to [43] for more details.

Table 1: Feasibility of Coin Tossing

Work	Setting	Security	Threshold	Broadcast	Feasible
Wu et al. [43]	GT	Comp.	$t \leq t^*, n \geq 3$ ⁶	✓	✓
Wu et al. [43]	GT	Comp.	$t \geq t^* + 1, n \geq 3$	✓	✗
Cleve [17]	Stand.	Comp.	$t \geq \frac{n}{2}, n \geq 2$	✓	✗
Cohen et al. [18]	Stand.	Comp.	$t \geq \frac{n}{3}, n \geq 3$	✗	✗
Rabin and Ben-Or [37]	Stand.	Stat.	$t < \frac{n}{2}, n \geq 2$	✓	✓
Ben-Or et al. [8]	Stand.	Stat.	$t < \frac{n}{3}, n \geq 3$	✗	✓
This work	GT	Stat.	$t \geq \frac{n}{2}, n \geq 2$	✓	✗
This work	GT	Comp.	$t \geq \frac{n}{3}, n \geq 3$	✗	✗

Our impossibility results (last two rows) can be seen as game-theoretic counterparts of the two highlighted results.

the flip side, Chung et al. proved that when precisely $n - 1$ parties are corrupt, unless all parties except for one have the same preference, CSP-fair coin toss is impossible to achieve. Wu, Asharov, and Shi [43] then extended this line of work by providing a complete characterization of game-theoretically fair, multi-party coin toss. Specifically, they proposed protocols in the setting where a majority ($\geq \frac{n}{2}$), yet less than $n - 1$, parties are corrupt, and complemented their positive results with matching lower bounds (see Table 1). Most recently, Thyagarajan, Soni, and Wu [41] extended this line of work by considering a multi-sided coin-toss in the same setting, and Zhang and Wu [44] explored the extension of this setting which accommodates arbitrary preference profiles.

We note, however, that the positive results of Wu et al. [43] and Thyagarajan et al. [41] fundamentally rely on computationally secure cryptographic tools in settings with fully malicious adversaries. For instance, Wu et al. employ secure multi-party computation with identifiable abort against a dishonest majority. However, under dishonest majority, MPC with such assumptions is known to be impossible to realize information-theoretically [31]. Consequently, these results do not provide game-theoretic substitutes for *information-theoretic* coin tossing in regimes where strong cryptographic guarantees are known to be impossible (e.g., dishonest-majority fairness).

Motivated by the success of utilizing game-theory to circumvent known cryptographic lower bounds, we thus ask the following natural question:

*Can game theory help us circumvent known lower bounds in the setting of **statistical** n -party coin-toss?*

In this work, we show that, perhaps surprisingly, the answer is **no**. In particular, we show that for $t \geq n/2$ corruptions, fair coin tossing is impossible in the statistical game-theoretic setting (with the exception of a corner case of $n = 4$ where three parties prefer the same bit), thereby complementing the existing positive results for $t < n/2$. Together, these establish a complete characterization of the feasibility landscape (with the exception of our corner case).

1.1 Our Results

Before presenting our formal results, we recall what is already known in the standard (non-game-theoretic) model. It is known [37] that in the standard, non-game-theoretic model, assuming broadcast and an **honest majority**, statistically secure multi-party computation with guaranteed output delivery, and hence, in particular, coin tossing with *strong* fairness, is achievable. Informally, strong fairness means that an adversary controlling up to t parties cannot predict or bias the outcome of the protocol toward *any* coin. Hence, it immediately implies the weaker, game-theoretic notions of fairness (e.g., CSP-fairness), since the adversary in particular cannot bias the outcome towards his preferred coin. In contrast, it is also known that **statistical coin tossing is impossible** when a *dishonest majority* of parties is allowed.

Our first result strengthens this impossibility by showing that, with the exception of one corner case, no statistically secure, **game-theoretic** coin-tossing protocol exists without an honest majority of parties. Formally, we prove the following.

Theorem. *For any fixed maximum number of rounds $\ell(\lambda)$, there exists no statistically secure, game-theoretic coin-toss protocol for n parties when $t \geq \lceil n/2 \rceil$ parties are corrupt, even under a broadcast channel, except for the special case $n = 4$, where all but one party share the same preference.*

Next, we turn our attention to the setting where we have an honest majority but at least a third is corrupt. By the above, we know that even a stronger primitive, like MPC, is possible in this setting *while assuming broadcast*. However, without broadcast (and without setup), the elegant result by Cohen, Haitner, Omri, and Rotem [18] shows that there is no non-trivial (i.e., achieving bias smaller than $\frac{1}{2}$) coin-tossing protocols for $t \geq n/3$, and $n \geq 3$.

As we show in this work, with the exception of one corner case, this lower bound extends to the game-theoretic setting, albeit with a different bias. In more detail, we establish the following.

Theorem. *In the secure point-to-point channel model (no broadcast), given $n \geq 3$ parties and an adversarial threshold $t \geq \frac{n}{3}$, there exists no computationally secure game-theoretic coin-tossing protocol with the round-complexity polynomial in λ , except for the special case $n = 6$, where all but one party share the same preference.*

Note that the above result gives an impossibility not only for the statistical, but also the computational setting. To obtain these results, we need to *extend the coin-tossing framework* of Chung et al. [16] to handle the *absence of broadcast*, which requires a careful treatment. Briefly, prior work on game-theoretic coin tossing typically assumed that the coin output is derived directly from messages sent over a broadcast channel. To capture settings without broadcast, we introduce revised utility and correctness definitions.

We note, however, that our impossibility result for the broadcast setting remains valid in the original model of Chung et al. [16] assuming that parties wish to output an unbiased random bit for any preference profile.⁷

We now describe our results in more detail.

2 Technical Overview

Game-theoretic coin-tossing protocols by Wu et al. [43] and Thyagarajan et al. [41] are built around the following simple yet powerful idea:

In the game-theoretic setting, when parties' preferences are publicly known, any party that misbehaves can be punished by setting the outcome to the opposite of its preferred value.

This idea is particularly compelling in the two-party case. Consider Blum's classical coin-toss protocol [11], and let Alice and Bob have public preferences $p_a, p_b \in \{0, 1\}$. Alice first commits to a random bit r_a by sending $\text{com}(r_a)$ to Bob. Bob then samples and publishes his own random bit r_b . Finally, Alice opens her commitment, and the coin is set to $c = r_a \oplus r_b$.

Note that after Bob published his random bit, Alice knows the outcome of the protocol, while Bob does not. In the standard setting, this allows Alice to bias the outcome by aborting after seeing r_b , simply by refusing to open her commitment. In the *game-theoretic* setting, however, such misbehavior can be deterred: if Alice refuses to open, the protocol instead outputs $c = 1 - p_a$, the value *opposite to Alice's preference*.

Such a penalty-based mechanism remains very useful in the *multi-party* setting, although applying it becomes more subtle. At a high level, even if a misbehaving party can be identified, a corrupt coalition may contain multiple parties with *different* preferences. Nevertheless, Wu et al. [43] and Thyagarajan et al. [41] obtain elegant constructions centered around the following ideas. They partition parties into groups according to their public preferences and then running an *MPC with identifiable abort* to toss a coin separately for each group (those preferring 0 and those preferring 1). Malicious parties can be gradually identified and excluded until the protocol reaches a point where either reconstruction succeeds, or – if reconstruction fails – the coalition's adversarial preference is implicitly revealed, allowing the protocol to set the output to the *opposite* of that preference.

While the availability of a *penalty-based approach* that leverages known preferences is powerful for constructing protocols, it makes proving *impossibility results* significantly more challenging. Consider, for example, the celebrated lower-bound by Cleve [17], which shows that in the standard (non-game-theoretic) setting an unbiased two-party coin-toss is impossible to achieve if an adversary is allowed to corrupt one of the two parties.

⁷ Chung et al. do not require parties to output each bit with probability $\frac{1}{2}$ if all parties' preferences are the same.

At a high level, Cleve’s argument defines $4r + 1$ adversarial strategies: the first simply aborts without sending the initial message, while each of the remaining strategies corresponds to a triple (**party**, **bit**, **message-num**). Here, **party** $\in \{\text{Alice}, \text{Bob}\}$ identifies the corrupted party, **bit** $\in \{0, 1\}$ is its preferred outcome, and **message-num** is the round after which the party computes the conditional outcome of the protocol. If that outcome disagrees with the preferred bit, the party aborts; otherwise it continues until the next round and may abort there. Cleve then shows that *at least one* of these strategies achieves a bias of $\Omega(\frac{1}{r})$.

This reasoning, however, **breaks down in the game-theoretic setting**: the fact that Alice can bias the output toward 1 does *not* imply that doing so increases her utility. In fact, if Alice prefers zero, she would actively harm herself by performing the attack.

We observe that the key in proving the impossibility of game-theoretic version of fairness is in showing that the adversary can bias the outcome of the protocol *towards his own preference*. Proving this is strictly harder than demonstrating (in line with traditional cryptographic literature) that the adversary can bias the outcome towards *some* value.

2.1 Impossibility of Statistical Game-Theoretic Coin-Toss Without Honest Majority

Establishing an analogue of Cleve’s lower bound in the *statistical game-theoretic* setting (note that Wu et al. [43] obtained positive results for certain dishonest-majority regimes in the game-theoretic *computational* setting; see also Table 1) requires new techniques, which we now introduce.

New tools for game-theoretic reasoning. Inspired by the elegant distributed computing literature which frequently models executions using graphs (e.g., Fischer, Lynch, and Paterson [25]), we introduce the notion of an *execution tree*. Classically, such graphs capture how a protocol evolves: nodes represent the internal states of the parties, and edges correspond to events that advance the system from one state to the next. In our setting, we adapt this idea to two-party protocols by letting each node represent a *partial transcript*, while each directed edge corresponds to one party sending a message and thereby extending the transcript of its parent node. Leaves (representing complete transcripts) are additionally annotated with the protocol’s outputs, as well as the probabilities with which each party outputs a given value. This structure will be a key tool in our impossibility arguments.

We further augment the notion of execution trees with the notion we call *black-pink root coloring*. Informally, for the execution tree of a coin-tossing protocol Π , we say the root is *black* if there exists an adversary corrupting Bob who can ensure that the outcome of the protocol matches Bob’s preferred bit at least with some probability $\delta(\lambda)$, where λ denotes the security parameter of the protocol. Similarly, the root is *pink* if there exists an adversary corrupting Alice who can ensure that the outcome of the protocol is Alice’s preferred bit at least with probability $\delta(\lambda)$. We call $\delta(\lambda) - \frac{1}{2}$ the *bias of the (black-pink) coloring*. If neither of the two applies, the root remains *colorless*. We believe that both the

execution-tree formalism and the black–pink root coloring may be useful beyond our specific results, especially for other works at the intersection of game theory and cryptography.

A note on the setting. In our presentation below, we consider the model in line with the original one defined by Chung et al. [16], which computes the coin outcome based only on the public broadcast messages. In our later impossibility of coin-tossing without broadcast, however, computing an outcome based on broadcast messages is not possible (given the absence of broadcast), hence in our model (Sec. 3) we require parties to compute the outcome based on their respective views, which in turn means that Alice and Bob may obtain different outputs. Our correctness definitions requires that the output mismatch should not happen except with a negligible probability, but nevertheless, this adjustment introduces a lot of subtleties in our formal proof of security (see Sec. 4).

Thus, for the ease of exposition, for the rest of this subsection we focus on the original model by Chung et al. [16]⁸. Briefly, it considers n parties with publicly known preferences encoded in a *preference profile*. A *preference profile* [16] is simply a vector of the parties’ preferences, e.g., preference profile $(0, 1)$ means that the first party prefers zero, while the second party prefers one. Parties can communicate with each other through pairwise private channels as well as a public broadcast channel. In this model, a coin-toss protocol is secure if a) all parties are honest, the protocol output is a uniform coin, and b) the protocol is *CSP-fair*, no adversarial coalition of t parties can increase its expected utility, irrespective of the its strategy. The utility of a coalition is a sum of utilities of its members, and a utility of each member is 1 if the protocol output matches the party’s preference, and 0 otherwise.

We now outline the intuition behind our proof of impossibility of a statistical game-theoretic coin-toss in the absence of honest majority. Towards this, we first consider the two-party setting, and distinguish between the preference profiles $(0, 0)$ and $(1, 1)$. We start with the $(0, 1)$ case.

Proving impossibility for the $(0, 1)$ -preference profile The core of our argument is to show that for any bound on the number of rounds in the protocol, the root of the execution tree of any given statistical two-party coin-toss for preference profile $(0, 1)$ can be colored either black or pink, where the coloring is for a bias $\frac{1}{2} - \text{negl}(\lambda)$. This, in turn, means that either Alice or Bob can ensure that the outcome of the protocol is their preferred bit with probability $1 - \text{negl}(\lambda)$.

We show this by induction on the number of rounds of the protocol, i.e., depth of the execution tree. Note that in our induction hypothesis we do not assume that the protocol is secure, we simply assume that it is a protocol for two parties which outputs either bit 0 or bit 1. For simplicity, say that in each round one party speaks. W.l.o.g., assume Alice is the first to send a message in the protocol (in this case, the only one). Intuitively, for single-round protocols, either there exists a message that Alice can send that results in the protocol

⁸ with the exception that we require parties to output an unbiased random bit even if all parties have the same preference

output being 0 (i.e., her preferred bit according to the preference profile), or all messages of Alice result in the protocol output being 1. In the first case, the root of the execution tree is pink, as Alice can simply choose a message leading to output zero. In the second case, the root is black, as Bob obtains his preferred outcome no matter what an honest Alice does.

To go from protocols with k rounds to those of $k + 1$, we note that removing the root (along with its incident edges) from the execution tree or a $(k + 1)$ -round protocol Π results in a forest of (sub)trees. Each of this subtrees can be seen as a k -round coin-toss protocol Π' (not necessarily a secure one), where Alice's first message of Π that lead to this subtree is encoded in the description of the new Π' , and otherwise the description of the protocol is exactly the same as that of Π (starting with its second round). By induction hypothesis, the roots of each of such subtrees is either black or pink. If there exists at least one pink root, Alice can bias the outcome in Π by choosing the message (corresponding to the execution tree's edge) leading to this subtree. Otherwise, if all roots are black, Bob can bias, as an honest Alice has no choice, but to take some edge leading to a black subtree, from which Bob has a strategy to bias.

We refer the reader to Section 4.2 for further details.

Proving impossibility for the $(1, 1)$ -preference profile First, note that in contrast to Chung et al. [16], which allow parties to simply output bit b if everyone prefers b , we require (perhaps more in line with traditional cryptographic literature) that parties output an unbiased random coin even in this setting.

To obtain the impossibility in this scenario we follow a similar approach of showing that the root of the execution tree of any given statistical two-party coin-toss for preference profile $(0, 1)$ can be colored either black or pink, however this time with a bias $\delta(\lambda)$ for some non-negligible λ . This, in turn, means that either Alice or Bob can ensure that the outcome of the protocol is their preferred bit with probability $\frac{1}{2} + \delta(\lambda)$.

While the proof structure is similar to the case of the $(0, 1)$ preference profile, there is a subtlety. Intuitively, as Alice and Bob both prefer the same bit, the intuition for the induction begin (i.e., protocols of one round) does not work anymore: If there exists no message of Alice that leads to her preferred outcome, it does *not* mean that all outcomes correspond to the preference of Bob anymore. We need to make use of the fact that the protocol is *correct*, i.e., outputs a fixed bit $b \in \{0, 1\}$ with probability at most $\frac{1}{2} + \text{negl}(\lambda)$. Now, we know that (informally) roughly half of Alice's messages lead to bit one, hence the root is pink.

Assuming the correctness of the protocol, however, makes the induction step trickier. Intuitively, the problem now is that in order to make use of the induction hypothesis, we need to show that the obtained subtrees correspond to *correct* k -round protocols. However, this does not immediately follow from the correctness of the $(k + 1)$ -round protocol Π : It could be the case that some of the trees are skewed toward, say, zero. Here, skewed towards zero means that an honest execution leads to an outcome zero with probability $\frac{1}{2} + \delta(\lambda)$ for some function δ that is non-negligible in λ . To solve this, we observe that if an honest execution of

Π leads to subtrees that are skewed towards zero non-negligibly often, there must be a subtree which is skewed towards one to “even” out the honest execution of Π . Clearly, as long as there exists one subtree which is skewed towards one, a malicious Alice can simply choose the message leading to such subtree, hence the root of Π can be colored pink. Otherwise, except with negligible probability, an honest execution of Π leads to subtrees which correspond to correct protocols. For these, we can apply our induction hypothesis and proceed by a similar argument as in the case the $(0, 1)$ -preference profile.

We refer the reader to Section 4.3 for further details.

Extending to multi-party setting. To prove impossibility in the general multi-party setting, we reduce to our previously shown two-party impossibilities. The key is to partition the parties into two groups of size at most $\frac{n}{2}$ each, such that either

- (a) one group prefers 0 and the other prefers 1 (allowing a reduction to the $(0, 1)$ -impossibility), or
- (b) both groups prefer 1 (allowing a reduction to the $(1, 1)$ -impossibility).

Above, a group prefers coin b if the (strict majority) of parties in this group prefer b . We refer the reader to Section 4.4 for details.

2.2 Impossibility of Game-Theoretic Coin-Toss for $t \geq \frac{n}{3}$ Without Broadcast

In our second result, we turn to the $t \geq \frac{n}{3}$ setting without broadcast. In the standard (non-game-theoretic) setting, Cohen et al. [18] showed that there exists no secure coin-toss protocol. Naturally, we asked ourselves whether it is possible to achieve statistically secure coin-toss in this setting while obtaining a game-theoretic fairness guarantee such as CSP-fairness (instead of the strong cryptographic one). However, it turns out that in this regime, it is impossible to achieve even computational protocols. Before we discuss our proof of impossibility, we discuss the implications of this new setting for our model.

Our (actual) model. While for the ease of exposition in the prior section we discussed the impossibility result assuming the underlying model of Chung et al. [16], in the setting without broadcast each party must compute its output based on *its own* view. Hence, while prior works assumed that the output is computed deterministically based on the public transcript, and the coin is thus implicitly the same for all parties, we must add an additional correctness requirement stating that the protocol’s output of the parties is the same (up to a negligible probability). To account for the fact that parties now have their own outputs, we furthermore have to re-define the utility of the adversarial coalition (otherwise an misbehaving party could “improve” its utility by simply outputting the preferred bit to himself). Instead, we set the utility of a coalition to be the fraction of *honest* parties’ outputs that match the adversarial preference, which intuitively corresponds to the idea that the adversary would like to bias the *honest parties*’s outputs towards his preference. As before, in contrast to Chung

et al. [16], we aim for an unbiased uniform coin-toss even if all parties have the same preference.

Proving impossibility for three parties, one corruption In order to show the impossibility of game-theoretic coin-toss in the $t \geq \frac{n}{3}$ setting without broadcast, where $n \geq 3$, we first outline the impossibility of a three-party coin-toss with one corruption. Here, our approach follows the elegant proof by Cohen et al. [18], albeit with adjustments to account for the game-theoretic setting. Intuitively, we consider an execution of a protocol which replicates parties of the original three-party one (which we assume secure) and places them in a ring, so that from the perspective of each party the execution could have been one in which the left party behaves honestly, or one in which the right party behaved honestly. We then show that in this new protocol, as long as the parties are not too many communication links apart, the parties output the same bit – which is furthermore an unbiased random one – except with a negligible probability. Next, we show an adversary who uses the emulation of this protocol to predict the outcome value of the original, real one, at the start of the execution. This will allow us to build an adversary who can bias the protocol’s output towards his preferred bit.

Later, we reduce the impossibility of a multi-party coin-toss where at least one third of parties can be corrupt, to that of the three-party case. See Section 5 for details.

3 Model and Security Definitions

We use the following model, partially borrowed from Wu et al. [43] and Thyagarajan et al. [41].

Network model. We consider n parties, denoted by P_1 to P_n , who interact through secure point-to-point channels as well as a public broadcast channel⁹. We assume that all communication channels are authenticated, i.e., messages always carry the true sender’s identity. We assume that the network is synchronous and the protocol proceeds in rounds.

Players’ preference profile. Each party has a publicly stated preference for either the bit 0 or the bit 1. We call the vector of all parties’ preferences the *preference profile* $(p_1, \dots, p_n)$, where p_i denotes the preference of the party P_i .

Adversary. We consider a computationally unbounded fully malicious adversary A that may corrupt up to t parties; the set of corrupted parties is also referred to as the *coalition*. The adversary A is allowed to perform a *rushing attack*: in each round r , the corrupted parties can wait to see all messages sent by honest parties (i.e., those not controlled by A) before (jointly) deciding which round- r messages to send.

Strategy. A strategy S_i of a party P_i describes the behavior of a party in each round of the protocol’s execution. A strategy S_I for a set I is the union of the strategies of all parties in I .

⁹ In Section 5 we adjust our model to point-to-point only, no broadcast

We let $H = \{H_1, \dots, H_n\}$ denote the strategy (the code) of the honest execution of protocol Π . That is, H_i is the code that party P_i is supposed to run according to the protocol specification. For a coalition A , we denote by $\bar{A}$ all parties not in A .

Utility. For a number k of honest parties, i.e., total number of parties minus the size of the coalition, let $(\text{out}_1, \dots, \text{out}_k)$ denote a vector of honest party's outputs. In line with prior works on game-theoretic coin toss [43,41] we consider the adversarial preference to be the preference of the majority of the parties in the coalition. Given A 's preference p , we define $\text{util}_A^* = 1 - \frac{\sum_{i \in [k]} |p - \text{out}_i|}{k}$. That is, if no honest party outputs the adversarial preference, util_A^* is zero. If every honest party outputs the adversarial preference, util_A^* of the adversary is one.

Given a strategy profile S_A for A and the honest strategy $H_{\bar{A}}$ for $\bar{A}$, the *expected utility* of A is

$$\text{util}_A(S_A, H_{\bar{A}}) := \mathbb{E}[\text{util}_A^*(\text{out}_1, \dots, \text{out}_k)],$$

where the expectation is over the parties' randomness when parties in A run S_A and parties in $\bar{A}$ run $H_{\bar{A}}$.

Similarly, $\text{util}_A(H_A, H_{\bar{A}})$ denotes the expected utility if everyone follows the honest strategy. In the following, we will refer to expected utility simply by "utility".

Coin toss definition. An n -party coin-toss protocol is an interactive protocol, where at the end of the protocol each party P_i computes the coin toss outcome out_i as a deterministic, polynomial-time function over the view of P_i , which includes its randomness r_i , as well as the set of messages sent and received by P_i (including public messages posted to the broadcast channel).

Definition 1. We say that the coin toss protocol Π , parametrized with a statistical security parameter λ , is **statistically secure** against any t -sized coalition with respect to a preference profile P , if the protocol provides correctness and cooperative-strategy-proofness (CSP-fairness), defined below:

- **Correctness.** If everyone behaves honestly, all parties output the same value out with probability at least $1 - \text{negl}(\lambda)$ for some negligible function $\text{negl}(\lambda)$. Further, out is some fixed $b \in \{0, 1\}$ with probability at most $1/2 + \text{negl}'(\lambda)$ for some negligible function $\text{negl}'(\cdot)$.
- **CSP-fairness.** For all adversaries A corrupting at most t parties, and any strategy S_A adopted by the coalition A , there is a negligible function $\text{negl}(\cdot)$, such that:

$$\text{util}_A(S_A, H_{\bar{A}}) \leq \text{util}_A(H_A, H_{\bar{A}}) + \text{negl}(\lambda)$$

Informally, the notion of CSP fairness requires that no coalition can increase its expected utility no matter how it deviates from the prescribed strategy. We further define **weak correctness** of a coin-toss protocol Π with a security parameter λ as follows:

- **Weak correctness.** If everyone behaves honestly, all parties output the same value out with probability at least $1 - \text{negl}(\lambda)$ for some negligible function $\text{negl}(\lambda)$.

We refer to the protocols that satisfy the (weak) correctness property as *(weakly) correct*, and to the protocols that satisfy CSP-fairness as *CSP-fair*.

4 Impossibility of Game-Theoretic Statistical Coin-Toss without Honest Majority

In this section we prove that a game-theoretic coin-toss is impossible to achieve in the information-theoretic setting without assuming honest majority. Towards this, we first introduce two notions that will be helpful in our impossibility results (Sec. 4.1). Next, we discuss two partial results: The impossibility of game-theoretic statistically-secure coin-toss in the two-party setting where one party is corrupt for the $(0, 1)$ -preference profile (Sec. 4.2), and for the $(1, 1)$ -preference profile (Sec. 4.3). Finally, we use our partial results to prove the impossibility of statistically-secure game-theoretic coin-toss in a setting without honest majority (Sec. 4.4).

4.1 Helper tools

Before proving our impossibility results, we introduce the following notion of *execution tree* of a two-party protocol Π that we will use throughout this section.

Execution tree. At a high level, an execution tree depicts all possible executions of the protocol between two honest parties. In more detail, w.l.o.g., assume exactly one party speaks per round and that the first party (Alice) speaks in odd rounds while the second party (Bob) speaks in even rounds. Let Π be a protocol between Alice and Bob in which each party sends at most K messages. Then:

Definition 2 (Execution tree). Let $\mathcal{R}_A$ and $\mathcal{R}_B$ denote the randomness spaces of Alice and Bob.

For a partial transcript $\tau = (m_1, \dots, m_\ell)$, define the honest support of the next message for a party $P \in \{A, B\}$ by

$$\text{Supp}_P(\tau) := \left\{ m \mid \exists (r_a, r_b) \in \mathcal{R}_A \times \mathcal{R}_B : \langle A(r_a), B(r_b) \rangle[1:\ell] = \tau \wedge \text{next}_P^r(\tau) = m \right\}.$$

Here $\langle A(r_a), B(r_b) \rangle[1:\ell]$ denotes the first ℓ messages of the honest execution of Π when Alice uses randomness r_a and Bob uses randomness r_b , and $\text{next}_P^r(\tau)$ is the next honest message party P produces after seeing τ when running with randomness r .

The execution tree of Π , denoted $\mathcal{T} = (V, E)$, is the rooted directed tree built as follows:

- The root corresponds to the empty transcript ε and is denoted v_ε .

- For each node v_τ already in V :
 - if it is Alice’s turn next (odd round), then for every $m \in \text{Supp}_A(\tau)$ add a child $v_{\tau \parallel m}$ and an edge $(v_\tau, v_{\tau \parallel m})$;
 - if it is Bob’s turn next (even round), then for every $m \in \text{Supp}_B(\tau)$ add a child $v_{\tau \parallel m}$ and an edge $(v_\tau, v_{\tau \parallel m})$.

Leaves (vertices with no outgoing edges) correspond to full protocol transcripts. For a leaf corresponding to transcript τ , we label it with the outputs and their respective probabilities for each party:

$$\begin{aligned} \text{SuppOut}(\tau) := \{ ((\text{out}_A, p_A), (\text{out}_B, p_B)) \mid \exists (r_A, r_B) \in \mathcal{R}_A \times \mathcal{R}_B : \langle A(r_A), B(r_B) \rangle = \tau \\ \wedge \text{Output}_A(r_A, \tau) = \text{out}_A \\ \wedge \text{Output}_B(r_B, \tau) = \text{out}_B \\ \wedge p_A = \Pr[\text{Output}_A = \text{out}_A \mid \tau] \\ \wedge p_B = \Pr[\text{Output}_B = \text{out}_B \mid \tau] \}, \end{aligned}$$

where Output_P denotes the output function of party P , and all probabilities are taken over the random tapes of Alice (resp., Bob) conditioned on the honest execution reaching τ .

Intuitively, every node of the tree correspond to a partial transcript between an honest Alice and an honest Bob up to a certain round, and the edge between two vertices corresponds to the message sent by the corresponding party that extends the transcript of the parent by precisely this message.

Black–pink root coloring. Additionally, we introduce the notion of a *black–pink root coloring*, which augments an execution tree by a coloring of its root. Specifically, the root is colored according to whether one of the parties can bias the protocol’s outcome toward its preferred value with probability at least p . Formally:

Definition 3 (Black–pink root coloring). Let $\mathcal{T} = (V, E)$ be the execution tree of a protocol Π as defined above. A black–pink root coloring with bias Δ augments $\mathcal{T}$ ’s root v_ϵ with a color according to a coloring function

$$c : \{v_\epsilon\} \longrightarrow \{\text{black}, \text{pink}, \perp\},$$

Where the following holds:

- $c(v_\epsilon) = \text{black}$ if an adversary that corrupts Bob at the very beginning of the protocol can ensure that Alice’s output of Π results in Bob’s preferred value with probability at least $\frac{1}{2} + \Delta$;
- $c(v_\epsilon) = \text{pink}$ if an adversary that corrupts Alice at the very beginning of the protocol can ensure that Bob’s output of Π results in Alice’s preferred value with probability at least $\frac{1}{2} + \Delta$.
- $c(v_\tau) = \perp$ otherwise.

Note that the distinction between Alice’s and Bob’s output is minimal: For correct protocols, these are the same except with a negligible probability.

In the following, we will say that a root of an execution tree $\mathcal{T}$ can be *colored black (resp., pink)* according to a coloring with bias Δ , if there exists a black-pink coloring of $\mathcal{T}$ with bias Δ in which the root has the specified color. If there exists no such coloring, we call the root colorless.

Finally, before discussing the impossibility results for specific preference profiles, we show that for weakly correct protocols (refer to Sec. 3 for the definition) with one message-sending round holds that honest executions almost always end up in leaves where both parties output the same value except with negligible probability. Formally:

Lemma 1. *Consider a weakly correct two-party coin-tossing protocol Π with security parameter λ consisting of a single message-sending round and an arbitrary preference profile. Then, except with negligible probability, any honest execution of Π finishes in a leaf τ which has a label*

$$((0, 1 - \text{negl}_1(\lambda)), (0, 1 - \text{negl}_2(\lambda))),$$

or a leaf which has a label

$$((1, 1 - \text{negl}_3(\lambda)), (1, 1 - \text{negl}_4(\lambda))),$$

for some functions $\text{negl}_1, \text{negl}_2, \text{negl}_3, \text{negl}_4$ that are negligible in λ .

Proof. W.l.o.g., Alice sends the first, and – since Π has a single message-sending round – only message of the protocol. Say there is a non-negligible probability that the honest execution ends up in leaves in which the following holds:

- For a non-negligible $\delta_0(\lambda)$, Bob’s output in this leaf is zero, and for (some other) non-negligible $\delta_1(\lambda)$, Bob’s output in this leaf is one.

Note that Alice has no knowledge of Bob’s randomness. Thus, for each leaf where the above holds Alice will fail to “guess” the output of Bob (in the sense that Alice’s output matches Bob’s) with a non-negligible probability. Hence, if the executions ends up in non-negligibly many such leaves, it breaks weak correctness.

Thus, except with negligible probability, honest executions end up in nodes in which either a) Bob’s output is zero with probability at least $1 - \text{negl}(\lambda)$, or b) Bob’s output is one with probability at least $1 - \text{negl}(\lambda)$ for some function negl that is negligible in λ . The lemma statement then follows by weak correctness of Π . $\square$

In the following, for the ease of notation, for $b \in \{0, 1\}$, we denote labels of the form

$$((b, 1 - \text{negl}_1(\lambda)), (b, 1 - \text{negl}_2(\lambda)))$$

simply by $(b, b, 1 - \text{negl}(\lambda))$.

4.2 Impossibility of Game-Theoretic Statistical Coin-Toss for $(0, 1)$ -Preference Profile and One Corruption

We now show our first result, namely that there exists no information-theoretic CSP-fair coin tossing protocol in the setting without honest majority. We start with a simpler result for two parties, and one corruption:

Theorem 1. *There exists no statistically secure information-theoretic coin-toss for $n = 2$, a preference profile $P = (0, 1)$, and a fixed maximum number of rounds $\ell(\lambda)$.*

Proof. Towards obtaining the result, we show the following helpful lemmas:

Lemma 2. *The root of the execution tree of any weakly correct information-theoretic two-party coin-toss protocol Π for the preference profile $(0, 1)$, which has a fixed maximum number of rounds ℓ , can be colored either black or pink according to a coloring with bias $\beta = \frac{1}{2} - \text{negl}(\lambda)$ for some function negl that is negligible in λ .*

Proof. We prove the result by induction on the number of rounds k . Our induction hypothesis is that there exists some function negl negligible in λ , so that consistent with the rules of the black-pink coloring with bias $\frac{1}{2} - \text{negl}(\lambda)$, the root of an execution tree of a weakly correct protocol Π , which has a fixed maximum number of rounds k , can be colored black or pink.

Say Alice prefers $b \in \{0, 1\}$ and Bob prefers $1 - b$.

Induction begin, $k = 1$. Consider a protocol Π of only round. W.l.o.g., Alice sends the first (and in this case only) message. By Lemma 1, we know that, except with negligible probability, an honest execution of Π ends up in leaves which have either the label $(0, 0, 1 - \text{negl}_0(\lambda))$, or the label $(1, 1, 1 - \text{negl}_1(\lambda))$. There are two cases to consider:

Case a. There exists a leaf with the label $(b, b, 1 - \text{negl}(\lambda))$. Therefore, Alice can bias Bob's output by publishing the message leading to such a leaf, and thus the root can be colored *pink*.

Case b. Otherwise, except with negligible probability, an honest execution ends up in leaves which have the label $(1 - b, 1 - b, 1 - \text{negl}(\lambda))$. Thus, except with negligible probability, Bob obtains his preferred outcome (i.e., Alice's output coincides with Bob's preference) regardless of Alice's first message, and the root can be colored *black*.

Induction step, $k \Rightarrow k + 1$. Consider a protocol Π of $k + 1$ rounds. If we remove the root and its incident edges from the protocol's execution tree, we get a forest $\mathcal{F}$ of (sub)trees of depth k . Note that each of these subtrees τ can be seen as an execution tree of a (possibly insecure) coin-toss protocol Π_τ with up to k rounds, where the message sent by Alice in the (now removed) edge of Π that lead to this subtree is hardcoded in Π_τ 's description, and for all rounds after the first one parties follow the instructions of the original Π . Say that some Π_τ that was defined in this way is not weakly correct, i.e., Alice's and Bob's outputs **do not**

coincide with some non-negligible probability $\delta(\lambda)$. As in Π this was the case at most with some negligible probability $\text{negl}'(\lambda)$, it means that an honest Alice chooses an edge leading to any of the subtrees that correspond to such Π_τ at most with some negligible probability $\text{negl}^*(\lambda)$.

For the remaining subtrees, by induction hypothesis, we get that the root of each such subtree can be colored either black or pink. We have two cases:

Case 1. If there is at least one root that is pink, in Π a malicious Alice can choose the message corresponding to the edge that leads to this subtree, and thus bias Bob's output towards her preference with the same probability as in this subtree. Hence, the root of Π 's execution tree is pink.

Case 2. Otherwise, if all roots of the remaining subtrees are black, in Π , once Alice sent her message, a malicious Bob can bias Alice's output towards his preference with the same probability as in the subtree that an honest Alice "chose" by sending her first message. By above, we get that an honest Alice chooses a subtree with a non-black root at most with probability $\text{negl}^*(\lambda)$, hence Bob can bias Alice's output towards his preference at least with probability $(1 - \text{negl}^*(\lambda)) * (1 - \text{negl}(\lambda)) = 1 - \text{negl}'(\lambda)$ for some negligible $\text{negl}'(\lambda)$. Hence, the root of Π 's execution tree is black.

□

Theorem 1 follows from Lemma 2 in a straight-forward way. Note that any correct protocol is also weakly correct. Thus, from Lemma 2 we get that the root of an execution tree of any correct protocol can be colored either black or pink according to a coloring with bias $\frac{1}{2} - \text{negl}(\lambda)$. By definition of the black-pink root coloring, we thus get that there exists an adversary who can ensure that the other party's output corresponds to the adversary's preference with probability at least $1 - \text{negl}(\lambda)$, thus violating CSP-fairness. Thus, we get that either a) the protocol does not satisfy correctness, or b) the protocol does not satisfy CSP-fairness. In both cases, the protocol is not secure. □

4.3 Impossibility of Game-Theoretic Statistical Coin-Toss for (1, 1)-Preference Profile and One Corruption

Theorem 2. *There exists no statistically secure information-theoretic coin-toss for $n = 2$, a preference profile $P = (1, 1)$, and a fixed maximum number of rounds ℓ , where all but one party share the same preference.*

Proof. Similar to the proof of Theorem 1, we first show the following lemma:

Lemma 3. *The root of the execution tree of any **correct** information-theoretic two-party coin-toss protocol Π for a preference profile $(1, 1)$, which has a fixed maximum number of rounds $\ell(\lambda)$, can be colored either black or pink according to a coloring with bias $\delta(\lambda)$ for some function δ that is non-negligible in λ .*

Proof. We prove the result by induction on the number of rounds k . Our induction hypothesis is that there exists a function δ that is non-negligible in λ , so

that consistent with the rules of the black-pink tree with bias $\delta(\lambda)$, the root of an execution tree of a correct protocol that has at most k rounds, can be colored black or pink.

Induction begin, $k = 1$. As every correct protocol is in particular weakly correct, we can apply the same reasoning as in the induction begin of the proof of Lemma 2, and get $\delta(\lambda) = \frac{1}{2} - \text{negl}(\lambda)$.

Induction step, $k \Rightarrow k + 1$. Consider a $(k + 1)$ -round correct protocol Π . If we remove the root and its incident edges from the protocol's execution tree, we get a forest $\mathcal{F}$ of (sub)trees of depth k . Note that each of these trees τ can be seen as an execution tree of a (possibly insecure) coin-toss protocol Π_τ with up to k rounds, where the message sent by Alice in the (now removed) edge of Π that lead to this subtree is hardcoded in Π_τ 's description, and for all rounds after the first one parties follow the instructions of the original Π . Say that for some Π' that was defined in this way Alice's and Bob's outputs **do not** coincide with some non-negligible probability. As in Π (by correctness) this was the case at most with some negligible probability $\text{negl}'(\lambda)$, it means that an honest Alice chooses an edge leading to any of the subtrees that correspond to such Π_τ at most with some negligible probability $\text{negl}^*(\lambda)$. We will account for this negligible probability further in the proof when giving a strategy for a malicious Bob (malicious Alice can simply chose the edge where outputs do coincide).

Consider the remaining subtrees, i.e., those in whose corresponding protocols Π_τ the outputs of Alice and Bob coincide except with negligible probability. Denote the forest of such subtrees by $\mathcal{F}'$. There are two cases:

Case 1. All of the subtrees in $\mathcal{F}'$ correspond to the protocols in which the output is a fixed $b^* \in \{0, 1\}$ with probability $\frac{1}{2} \pm \text{negl}(\lambda)$ for some negligible function $\text{negl}(\lambda)$.

In this case, all such protocols are correct, and, as they have at most k rounds, by induction hypothesis we get that the root of every subtree $\in \mathcal{F}'$ is either black or pink. If there is at least one root that is pink, in Π a malicious Alice can choose the message corresponding to the edge that leads to this subtree, and thus bias Bob's output towards her preference with the same probability as in this subtree. Hence, the root of Π 's execution tree is pink. Otherwise, if all roots of the remaining subtrees are black, in Π , once Alice sent her message, a malicious Bob can bias Alice's output towards his preference with the same probability as in the subtree that an honest Alice "chose" by sending her first message. By above, we get that an honest Alice chooses a subtree with a non-black root at most with probability $\text{negl}^*(\lambda)$, hence by induction hypothesis Bob can bias Alice's output towards his preference at least with probability $(1 - \text{negl}^*(\lambda))(\frac{1}{2} + \delta(\lambda)) = \frac{1}{2} + \delta'(\lambda)$, for some δ' non-negligible in λ .¹⁰ Hence, the root of Π 's execution tree is black.

Case 2. There exists a subtree in $\mathcal{F}'$ in whose corresponding protocol Π' the proportion of 1 and 0 outputs is skewed, i.e., there exists a non-negligible

¹⁰ Note that by Bernoulli's inequality $(1 - \text{negl}(\lambda))^{\text{poly}(\lambda)} \geq 1 - \text{poly}(\lambda) * \text{negl}(\lambda)$, hence we can safely repeat this step for polynomial-depth protocols.

$\delta'(\lambda)$ such that in the corresponding subtree the protocol output under honest execution is some fixed b with probability $\frac{1}{2} + \delta'(\lambda)$.

If such subtree exists for $b = 1$, a malicious Alice can always choose this subtree, in which case we can color the root of Π pink.

Otherwise, as the sum of probabilities that Alice chooses a given first edge sums up to 1, if for all “skewed” subtrees $b = 0$, it must be the case that an honest Alice chooses edges leading to the subtrees that correspond to such Π ’s at most with some negligible probability $\text{negl}^{**}(\lambda)$ (otherwise Π cannot be correct). In this case, as all remaining subtrees correspond to the protocols in which the output is a fixed $b^* \in \{0, 1\}$ with probability $\frac{1}{2} \pm \text{negl}(\lambda)$ for some negligible function $\text{negl}(\lambda)$, by induction hypothesis we get that their roots are either black or pink. Hence, by a similar argument as before, we get that Π ’s root is also black or pink.

□

Now, from Lemma 3 we get that the root of an execution tree of any correct protocol can be colored either black or pink according to a coloring with bias $\delta(\lambda)$ for some function δ that is non-negligible in λ . By definition of the black-pink root coloring, we obtain that there exists an adversary who can ensure that the other party’s output corresponds to the adversary’s preference with probability at least $\frac{1}{2} + \delta(\lambda)$, which violates CSP-fairness. Thus, we get that either a) the protocol does not satisfy correctness, or b) the protocol does not satisfy CSP-fairness. In both cases, the protocol is not secure. Hence, Theorem 2 follows. □

4.4 Impossibility of Game-Theoretic Statistical Coin-Toss without Honest Majority

We now finalize our result and show that without honest majority there exists no game-theoretic statistically secure for an arbitrary number of parties.

Theorem 3. *There exists no statistically secure information-theoretic coin-toss for a fixed maximum number of rounds $\ell(\lambda)$, n parties, and $t \geq \lceil n/2 \rceil$ corruptions, except for the special case $n = 4$, where all but one party share the same preference.*

Proof. Let n_0 denote the number of parties who prefer zero, and let $n_1 = n - n_0$ denote the number of parties who prefer one. W.l.o.g. we assume $n_1 \geq n_0$. There are three cases:

- **Case 1:** $n_0 < \frac{n}{4}$.
- **Case 2:** $n_0 = \frac{n}{4}$.
- **Case 3:** $\frac{n}{4} < n_0 < \frac{n}{2}$.

We now discuss each of these cases in detail.

Case 1. Say there exists a game-theoretic statistically secure coin-toss protocol Π for n parties, where less than $\frac{n}{4}$ parties prefer zero. Split the parties into

a group **(a)** of $\lceil \frac{n}{2} \rceil$ parties who prefer one, and **(b)** remaining parties. Given Π , we can construct a protocol Π' for the two-party, one corruption case with preference profile $(1, 1)$ as follows.

We let Alice execute (next-message functions of) protocol Π for the parties of group **(a)**, while Bob executes protocol Π for the parties of group **(b)**. In the end, Alice takes as output any output produced by a party of group **(a)**, while Bob takes as output any output produced by a party from group **(b)**.

Note that both groups are of size at most $\lceil n/2 \rceil$. Further, note that as $n_0 < \frac{n}{4}$, we get that in group **(b)** more parties prefer one than zero. As Π is secure against adversaries corrupting up to $\lceil n/2 \rceil$ messages, by CSP-fairness of Π we get that neither group **(a)**, nor group **(b)** can bias the output of the protocol towards one. Thus, we get that Π' satisfies CSP-fairness for the preference profile $(1, 1)$. By correctness of Π we further get that up to a negligible probability, the output of all parties is the same value, and this value is a bit $b \in \{0, 1\}$ with probability at most $\frac{1}{2} + \text{negl}'(\lambda)$. Thus, we get that Π' satisfies correctness. Together, we obtain that Π' is a statistically secure game-theoretic coin-toss protocol for the preference profile $(1, 1)$ and one corruption - a contradiction to Theorem 2.

Case 2.

First, we consider the general case: Say there exists a game-theoretic statistically secure coin-toss protocol Π for $n > 4$ where $n_0 = \frac{n}{4}$. We split the parties into a group **(a)** of $\frac{n}{2} - 1$ parties who prefer one along with one party who prefers zero, and **(b)** remaining parties. As $\frac{n}{2} - 1 > 1$, group **(a)** prefers one. As $n - (\frac{n}{2} - 1) - n_0 = n_0 + 1 > n_0 - 1$, group **(b)** also prefers one. From here we proceed the same as in **Case 1**, and obtain a contradiction to Theorem 2.

Now, consider the special case with four parties where three parties prefer one. The problem in this scenario is that, no matter how we split parties into groups of size at most two, there exists one group with exactly one party preferring one, and one party preferring zero. Thus, the preference of this group is undefined, and we cannot reduce it to our two-party impossibility. We leave the exploration of this preference profile as an interesting direction for future work.

Case 3. Say there exists a game-theoretic statistically secure coin-toss protocol Π for n parties, where over $\frac{n}{4}$, yet less than $\frac{n}{2}$ parties prefer zero. Split the parties into a group **(a)** of $\lceil \frac{n}{2} \rceil$ parties who prefer one, and **(b)** remaining parties. Given Π , we can construct a protocol Π' for the two-party, one corruption case with preference profile $(0, 1)$ as follows.

We let Alice execute (next-message functions of) protocol Π for the parties of group **(a)**, while Bob executes protocol Π for the parties of group **(b)**. In the end, Alice takes as output any output produced by a party of group **(a)**, while Bob takes as output any output produced by a party from group **(b)**.

Note that both groups are of size at most $\lceil n/2 \rceil$. Further, as $n_0 > \frac{n}{4}$, in group **(b)** more parties prefer zero than one. As Π is secure against adversaries corrupting up to $\lceil n/2 \rceil$ messages, by CSP-fairness of Π we get that group **(a)** cannot bias the output towards one, while group **(b)** can bias the output of the protocol towards zero. Thus, we get that Π' satisfies CSP-fairness for the preference profile $(0, 1)$. By correctness of Π we further get that up to a neg-

ligible probability, the output of all parties is the same value, and this value is a bit $b \in \{0, 1\}$ with probability at most $\frac{1}{2} + \text{negl}'(\lambda)$. Thus, we get that Π' satisfies correctness. Together, we obtain that Π' is a statistically secure game-theoretic coin-toss protocol for the preference profile $(0, 1)$ and one corruption - a contradiction to Theorem 1. $\square$

5 Impossibility of Game-Theoretic Coin-Toss for $n \leq 3t$ Without Broadcast

In the standard (non-game-theoretic) setting in the secure point-to-point channel model (without setup and without broadcast) an elegant result by Cohen et al. [18] shows that there exist no non-trivial (i.e., bias smaller than $\frac{1}{2}$) coin-tossing protocols for $n \leq 3t$, $n \geq 3$. We now show that this result extends to the game-theoretic setting, albeit with a different bias. As this result holds even against computational adversaries, we first define the computational version of our coin-tossing notion:

Coin-tossing, computational. As before, for a protocol Π , we let H denote the strategy of the honest execution. For a coalition A , we denote by $\bar{A}$ all parties not in A . We denote by $\text{util}_A(S_A, H_{\bar{A}})$ the expected utility of all members in A where the members of A follow some strategy S_A , and the parties who are not in A follow the honest strategy $H_{\bar{A}}$. Similarly, $\text{util}_A(H_A, H_{\bar{A}})$ denotes the expected utility of all members in A where both the members of the coalition A , and the parties who are not in A , follow the honest strategy.

Definition 4. We say that the coin-toss protocol Π , parametrized with a security parameter λ , is **computationally secure** against any t -sized coalition with respect to a preference profile P , if the protocol provides correctness and cooperative-strategy-proofness (CSP-fairness), defined below:

- **Correctness** If everyone behaves honestly, all parties output the same value out with probability at least $1 - \text{negl}(\lambda)$ for some negligible function $\text{negl}(\lambda)$. Further, out is some fixed $b \in \{0, 1\}$ with probability at most $1/2 + \text{negl}(\lambda)$.
- **CSP-fairness** For all PPT adversaries A corrupting at most t parties, and any PPT strategy S_A adopted by the coalition A , there is a negligible function $\text{negl}(\cdot)$, such that:

$$\text{util}_A(S_A, H_{\bar{A}}) \leq \text{util}_A(H_A, H_{\bar{A}}) + \text{negl}(\lambda)$$

Theorem 4. In the secure point-to-point channel model (no broadcast), given $n \geq 3$ parties and an adversarial threshold $t \geq \frac{n}{3}$, there exists no computationally secure coin-toss with round-complexity that is polynomial in λ , except for the special case $n = 6$, where all but one party share the same preference.

To show this, we adapt the elegant proof by Cohen et al. [18] to the game-theoretic coin-tossing setting.

First, we show the result for three parties:

Theorem 5. *In the secure point-to-point channel model (no broadcast), given $n = 3$ parties and adversarial threshold $t = 1$, for any preference profile, and any maximum number of protocol rounds q which is polynomial in λ , there exists a PPT adversary A that can corrupt any party P_i , and ensure that with probability at least $\frac{3}{4} - \text{negl}(\lambda)$, the output of every honest party is P_i 's preference p_i .*

Proof. For the sake of contradiction, assume there exists a computationally secure q -round protocol Π for three parties. Let A , B , and C denote the code executed by each of the three parties in Π . We construct a new protocol Π^* for $12q$ parties as follows. Let the parties be $P_1, \dots, P_{12q}$. The parties are arranged in a ring: each P_i is connected to P_{i-1} and P_{i+1} , where indices are taken modulo $12q$, and P_1 is connected to P_{12q} and P_2 , and P_{12q} is connected to P_{12q-1} and P_1 . We define

$$P_i \text{ executes } \begin{cases} A & \text{if } i \bmod 3 = 1, \\ B & \text{if } i \bmod 3 = 2, \\ C & \text{if } i \bmod 3 = 0. \end{cases}$$

In the following, we denote this arrangement as

$$\Pi' = (A^1, B^1, C^1, \dots, A^{4q}, B^{4q}, C^{4q}).$$

First, we show the following helpful lemma, where distance d between the two parties is the minimal possible number of communication links between them:

Lemma 4. *In an execution of Π^* , where parties have an arbitrary preference profile, and each party's random tape is chosen uniformly at random, any two parties within distance d and every party between them output the same value with probability at least $1 - d \cdot \text{negl}(\lambda)$, and further this output is some fixed $b \in \{0, 1\}$ with probability at most $1/2 + \text{negl}(\lambda)$.*

Proof. First, for any preference profile and any pair of neighboring parties A^j, B^j , we consider an execution of Π , where the adversary A controls party C . In this execution, A internally emulates the execution of Π^* , where the emulated parties A^j, B^j interact exactly as A and B of Π (i.e., upon obtaining a message from A , A emulates A^j sending this message to C^{j-1} , upon obtaining a message from B , A emulates B^j sending this message to C^j , upon C^{j-1} sending a message to A^j , A sends the same message to A , and upon C^j sending a message to B^j , A sends the same message to B). As A corrupts only one party, and Π is a computationally secure coin-tossing protocol, by correctness we get that A and B output the same bit except with a negligible error probability $\text{negl}(\lambda)$. Further, as the joint view of A^j, B^j has the same distribution as that of A, B , we get that A^j and B^j output the same value with the same probability as A and B do, i.e., $1 - \text{negl}(\lambda)$, and (by correctness of Π) this output is some fixed $b \in \{0, 1\}$ with probability at most $1/2 + \text{negl}(\lambda)$.

Note that the argument above works analogously for any other pair of neighboring parties, e.g., C^{j-1} together with A^j , and B^j together with C^j . Thus, by union bound we get that any parties within distance d (and everyone in-between), outputs the same value with probability at least $1 - d \cdot \text{negl}(\lambda)$. $\square$

Now, for any i , and an arbitrary preference profile, we construct an adversary A for Theorem 5 as follows. A corrupts a party $P_i \in \{A, B, C\}$, and emulates an execution of Π^* , sampling the randomness tapes of all parties in Π^* uniformly at random. If the output of party A^{2q} **does not** match the preference of P_i , A simply continues the execution honestly. Otherwise, A proceeds as we outline below. By Lemma 4, we get that the probability of failure is at most $\frac{1}{2} + \text{negl}'(\lambda)$.

Next, in its interaction with the honest parties of Π , the adversary emulates the execution of Π^* , so that the honest parties of Π correspond to B^1 and C^1 (if P_i is A), to A^1 and C^1 (if P_i is B), and to A^1 and B^1 (if P_i is C). That is, whenever an honest party P_j in Π sends a message, A forwards it to the corresponding neighbor of the party acting as P_j in Π^* , and vice versa. In this execution, A sets the randomness tapes of all parties (except those acting as the two honest parties of Π) to match the randomness previously used in the execution in which A^{2q} output P_i 's preferred value. We make the following observations:

First, the view of A^{2q} in this attack is the *same* as its view in the execution of Π on the original random tapes that led to the output y . This is because the only random tapes that we changed between the two executions are those of the parties who act as their Π counterparts, these parties are at least q communication links away from A^{2q} , and the execution of Π^* *finishes* after at most q rounds.

Second, note that in the execution of Π^* in which A^{2q} 's output coincided with P_i 's preference we set the random tapes of parties uniformly at random, and the distance between A^{2q} and the two parties who act as their Π counterparts is at most $6q$. Thus, by Lemma 4 together with our first observation we get that with probability $1 - 6q \cdot \text{negl}(\lambda)$, the output of the honest parties in Π is the same as that of A^{2q} , i.e., P_i 's preference p_i .

Note that if A^{2q} 's output did not coincide with P_i 's preference, as the adversary next behaved honestly in Π , by correctness the protocol's output matched the adversarial preference with probability $\frac{1}{2} - \text{negl}(\lambda)$. All in all, we get that our adversary succeeds with probability at least $(\frac{1}{2} - \text{negl}(\lambda)) \cdot (\frac{1}{2} - \text{negl}'(\lambda)) + (\frac{1}{2} - \text{negl}'(\lambda)) \cdot (1 - 6q \cdot \text{negl}(\lambda)) = \frac{3}{4} - \text{negl}^*(\lambda)$ for some negligible function negl^* . $\square$

Clearly, the existence of adversary A from Lemma 5 implies the impossibility of a three-party, one corruption coin-toss in the setting without broadcast *for any given preference profile*. We now use this result to prove Theorem 4.

Proof. Consider an arbitrary preference profile p , and let n_0 denote the number of parties who prefer zero, and $n_1 = n - n_0$ denote the number of parties who prefer one. W.l.o.g., say $n_1 \geq n_0$, and let $k = n_1 - \lceil \frac{n}{3} \rceil$. We have the following cases:

- **Case 1:** $k \geq n_0$
- **Case 2:** $k < n_0$

Case 1. For the sake of contradiction, say there exists a game-theoretic coin-toss protocol for $n \geq 1$ parties, adversarial threshold $t \geq \frac{n}{3}$ and $k \geq n_0$. Then,

if **(case 1a.)** $k - \lceil \frac{n}{3} \rceil \neq n_0$, we can split the parties into the following three groups: **(1)** $\lceil \frac{n}{3} \rceil$ parties who prefer one, **(2)** $\lceil \frac{n}{3} \rceil$ parties who prefer one, and **(3)** remaining parties. We reduce to the impossibility of the 3-party coin-toss for the preference profile $(1, 1, b)$, where b is the preference of the majority of parties in group **(3)**. Note that b is well-defined, i.e., there is a strict majority as $k - \lceil \frac{n}{3} \rceil \neq n_0$. The reduction is as follows.

Say there exists a protocol Π for n parties which can be split into the three groups as outlined above. We construct a protocol Π' for a three-party coin-toss with the preference profile $(1, 1, b)$ as follows. Call the three parties Alice, Bob, and Charlie. We let Alice execute protocol Π for all parties of group **(1)** and take as output any output value of this group. Bob does the same with group **(2)**, and Charlie with group **(3)**.

Note that all groups are at most size $\frac{n}{3}$. Further, the majority preference of groups **(1)** and **(2)** is one, while the majority preference of group **(3)** is b . By CSP-fairness of Π neither group **(1)** nor group **(2)** can bias the protocol outcome of the honest parties towards one, and group **(3)** cannot bias the honest outcomes toward b . Further, by correctness of Π , all parties output the same bit which is furthermore some fixed $b^* \in \{0, 1\}$ with probability at most $\frac{1}{2} + \text{negl}(\lambda)$. Thus, we get that Π' is secure three-party coin-toss protocol for the preference profile $(1, 1, b)$, hence we reached a contradiction.

Otherwise, if **(case 1b.)** $k - \lceil \frac{n}{3} \rceil = n_0$ and further $n \neq 6$, we split the parties into the following three groups: **(1)** $\lceil \frac{n}{3} \rceil$ parties who prefer one, **(2)** $\lceil \frac{n}{3} \rceil - 1$ parties who prefer one plus one party who prefers zero, and **(3)** remaining parties. We reduce to the impossibility of the 3-party coin-toss for the preference profile $(1, 1, 1)$ by the same argument as above. Otherwise, it remains that $n = 6$ and $k - \lceil \frac{n}{3} \rceil = n_0$, i.e., exactly five parties have preference one. Note that the problem in this scenario is that, no matter how we split parties into groups of size at most two, there exists one group with exactly one party preferring one, and one party preferring zero. Thus, the preference of this group is undefined, and we cannot reduce it to our three-party impossibility. We leave the exploration of this preference profile as an interesting direction for future work.

Case 2. For the sake of contradiction, say there exists a game-theoretic coin-toss protocol for $n \geq 1$ parties, adversarial threshold $t \geq \frac{n}{3}$ and $k < n_0$. Then, if **(case 2a.)** $n_0 - \lceil \frac{n}{3} \rceil \neq k$, we can split the parties into the following three groups: **(1)** $\lceil \frac{n}{3} \rceil$ parties who prefer one, **(2)** $\lceil \frac{n}{3} \rceil$ parties who prefer zero, and **(3)** remaining parties. We reduce to the impossibility of the 3-party coin-toss for the preference profile $(1, 0, b)$, where b is the preference of the majority of parties in group **(3)**. Note that b is well-defined, i.e., there is a strict majority as $n_0 - \lceil \frac{n}{3} \rceil \neq k$.

If **(case 2b.)** $n_0 - \lceil \frac{n}{3} \rceil = k$, and $n \neq 6$, we split the parties into the following three groups: **(1)** $\lceil \frac{n}{3} \rceil$ parties who prefer one, **(2)** $\lceil \frac{n}{3} \rceil - 1$ parties who prefer zero plus one party who prefers one, and **(3)** remaining parties. We reduce to the impossibility of the 3-party coin-toss for the preference profile $(1, 0, 1)$. Otherwise, it remains that $n_0 - \lceil \frac{n}{3} \rceil = k$ and $n = 6$, i.e., exactly five parties have

preference zero. As above, we leave a detailed study of this preference profile as an interesting avenue for future work. $\square$

6 Conclusion

In this work, we proved two new impossibility results for game-theoretically secure coin tossing. First, we showed that even *assuming broadcast*, a statistically secure, game-theoretic coin-toss protocol is impossible to achieve without an honest majority. Second, we proved that *without broadcast*, once at least one third of the parties may be corrupt, no game-theoretically secure coin toss is possible—even with only computational security. Along the way, we extended the model and security notions of Chung et al. [16] to reason formally about settings *without broadcast*. We further introduced the concepts of *execution trees* and *black–pink root coloring*, which may be of independent interest.

Taken together with known results from the cryptographic literature, our findings yield an (almost) complete picture of the feasibility of *statistical, game-theoretic coin tossing*:

- **With broadcast but without an honest majority:** we show impossibility for all preference profiles, except for the special case $n = 4$ when exactly three parties share the same preference.
- **With broadcast and an honest majority:** strongly fair (and therefore also game-theoretically fair) coin tossing follows from classical results (e.g., Rabin and Ben-Or [37]).
- **Without broadcast and without a strict two-thirds honest majority:** we prove impossibility for all preference profiles, except for the special case $n = 6$ when exactly five parties share the same preference.
- **Without broadcast but with a strict two-thirds honest majority:** strongly fair, statistically secure coin tossing is once again possible (see, e.g., Ben-Or et al. [8]).

We leave the exploration of the two corner cases, i.e., $n = 4$ when exactly three parties share the same preference and $n = 6$ when exactly five parties share the same preference, as an interesting direction for future work.

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